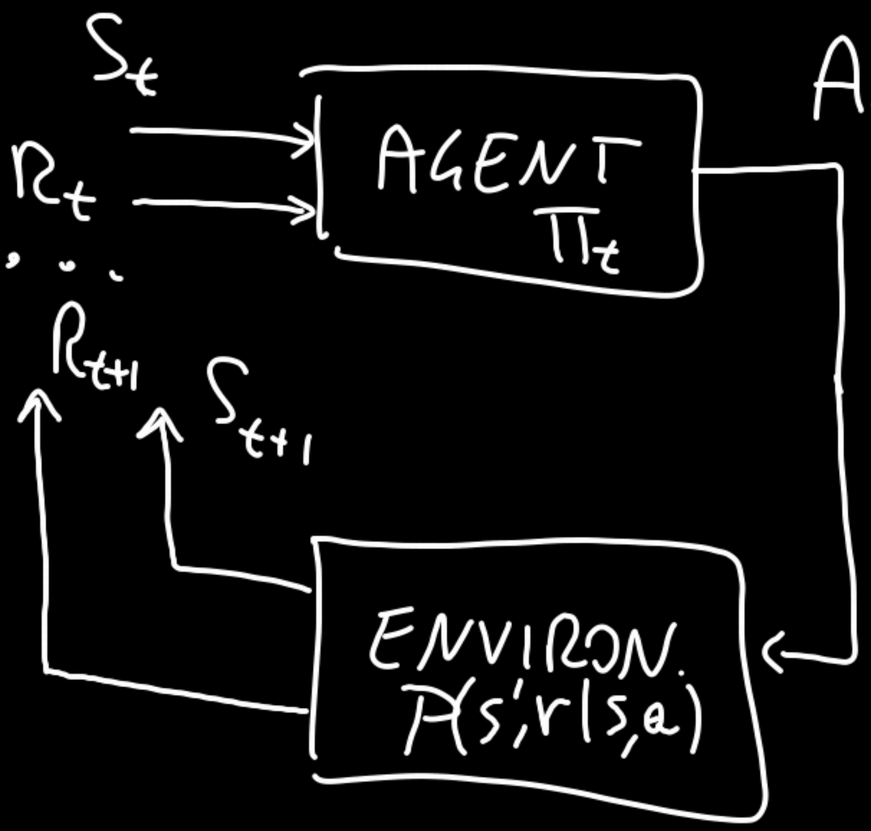
6/11/2026 - BELLMANN EQUATION → DERIVATION OF OPTIMAL POLICIES IN CASE OF KNOWN ENVIRONMENT (DYNAMIC PROGRAMMING).



MODEL OF ENVIRONMENT

$$p(s',r|s,a) \triangleq \Pr\{S_t = s', R_t = r \mid S_{t-1} = s, A_{t-1} = a\}$$

$$S_0, A_0, R_1, S_1, A_1, R_2, S_2, \ldots$$

DISCOUNTED RETURN ($\gamma \in [0,1]$)

$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1} = 1R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \cdots$$

VALUE FUNCTIONS

$v_\pi(s) : S \rightarrow \mathbb{R}$ EXPECTED DISCOUNTED RETURN FROM s, FOLLOWING π.

↳ STATE VALUE FUNCTION

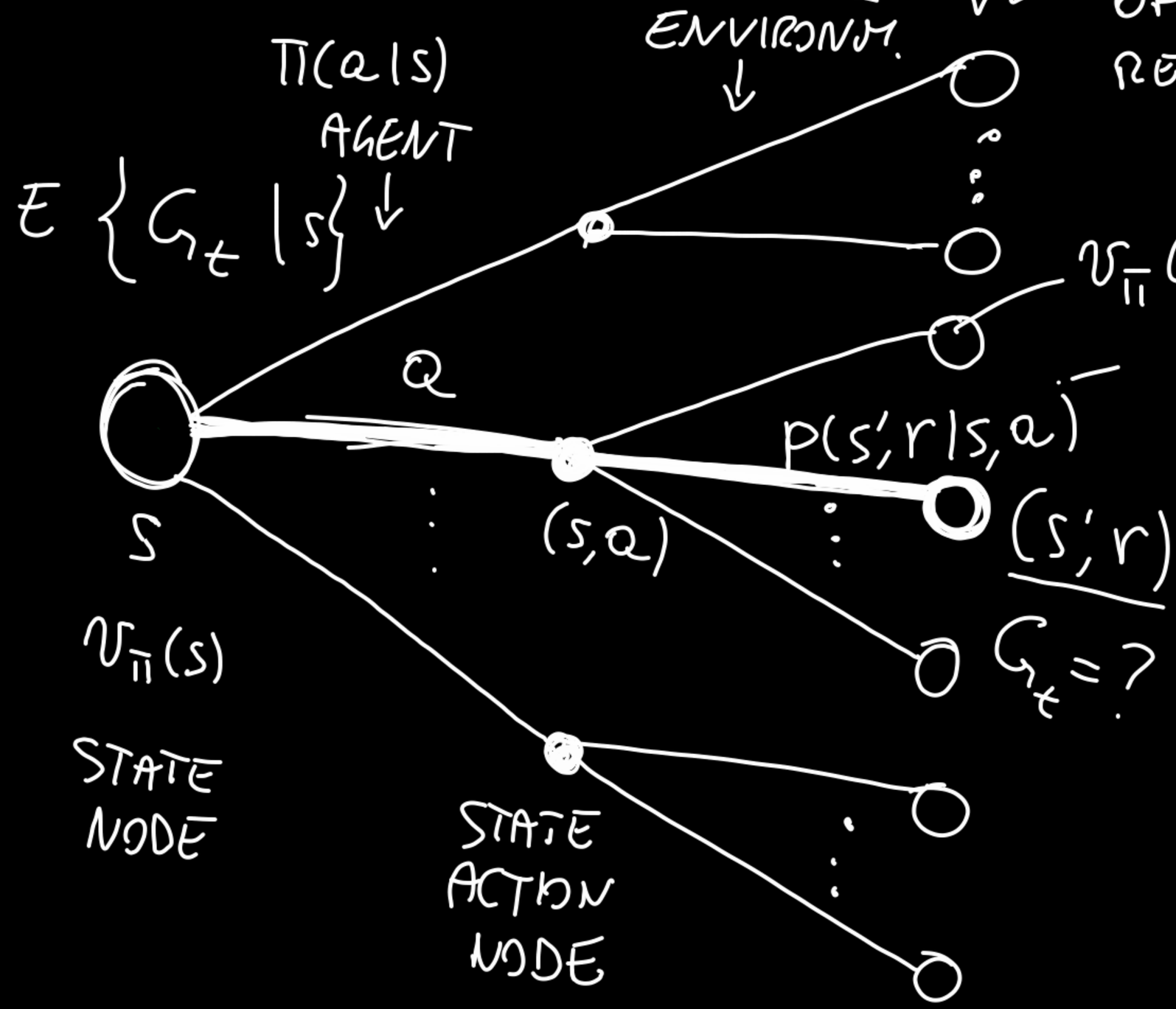
$$v_\pi(s) \triangleq E_\pi\{ G_t \mid S_t = s \}$$

$q_\pi(s,a)$ STATE-ACTION VALUE FUNCTION.

$$q_\pi(s,a) = E_\pi\{ G_t \mid S_t = s, A_t = a \}$$

$$\boxed{G_t = R_{t+1} + \gamma G_{t+1}}$$ ITERATIVE RELATION ON G_t

BELLMAN EQUATION ON $v_\pi(s)$



① COMPUTE THE EXPECTED RETURN FOR EVERY POSSIBLE NODE AT SECOND LEVEL ((s',r)).

② PROBABILITY OF ARRIVING AT A SPECIFIC COMBINATION (s',r)

$$v_\pi(s) = E\{G_t \mid S_t = s\} =$$

G_t $\quad$ $\pi(a|s)$ $\quad$ $p(s', r \mid s, a)$



$$v_\pi(s) = r + E\{\gamma G_{t+1} \mid S_{t+1} = s'\}$$

$$v_\pi(s) = E\{R_{t+1} + \gamma G_{t+1} \mid S_t = s\}$$

$$\boxed{v_\pi(s) = E\{G_t \mid S_t = s\}}$$

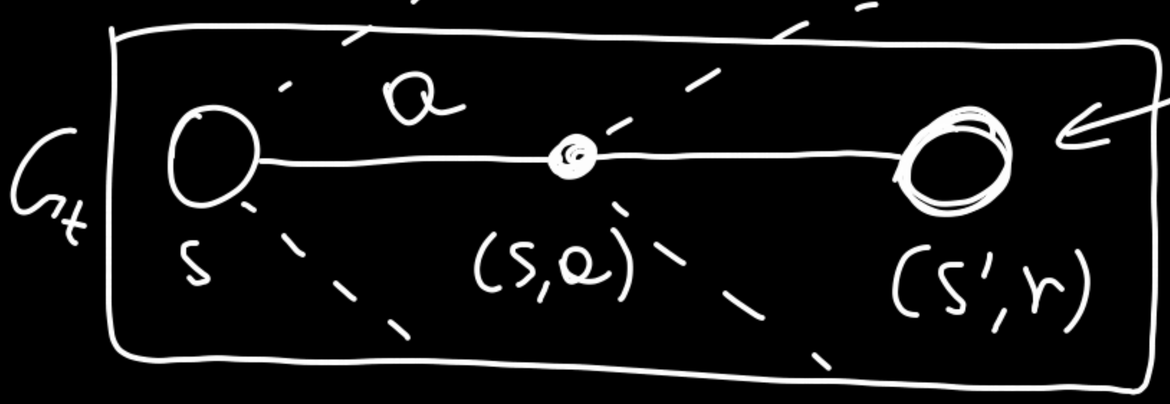
$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \cdots = R_{t+1} + \gamma G_{t+1}$$

($R_{t+1} \leftarrow r$, $G_{t+1} \leftarrow S_{t+1} = s'$)

$$\rightarrow v_\pi(s) = r + \gamma \boxed{E\{G_{t+1} \mid S_{t+1} = s'\}} = r + \gamma v_\pi(s')$$

DEFINITION OF v_π

FOR A PARTICULAR TRANSITION



→ PROBABILITY OF $\pi(a|s) \cdot p(s', r|s, a)$

WE FOUND

$$v_\pi(s) = r + \gamma v_\pi(s')$$

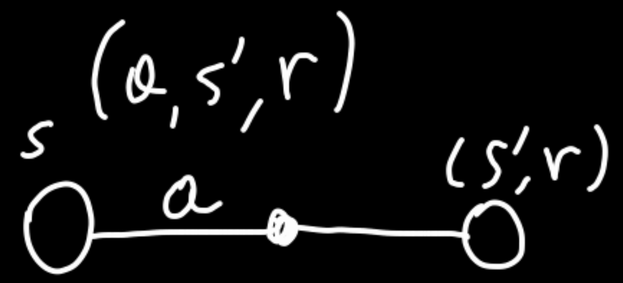
$r + \gamma v_\pi(s')$ IS THE VALUE OF THE EXPECTED RETURN FROM S, WHEN THE GENERIC ACTION a IS TAKEN, AND THE SPECIFIC NEXT STATE S' AND REWARD r ARE OBSERVED/OBTAINED.

NOW WE AVERAGE OVER ALL POSSIBLE CASES

BELLMAN EQUATION:

$$v_\pi(s) = \sum_{\text{ALL POSSIBLE CASES}} \left(\begin{array}{l}\text{EXPECTED RETURN}\\ \text{IN THAT CASE}\end{array}\right) \cdot \left(\begin{array}{l}\text{PROBABILITY}\\ \text{OF OCCURRENCE}\\ \text{OF THE CASE}\end{array}\right)$$

$$v_\pi(s) = \underbrace{\sum_a \sum_{s'} \sum_r}_{\text{EXPLORING ALL POSSIBLE CASES (COMBINATIONS OF } a, s', r)} \underbrace{\left[\pi(a|s) \cdot p(s', r|s, a)\right]}_{\text{PROBABILITY OF } (a, s', r)} \cdot \underbrace{\left[r + \gamma v_\pi(s')\right]}_{\text{EXPECTED RETURN IN THAT CASE}}$$



BELLMAN EQ. FOR $v_\pi(s)$

KNOWN KNOWN

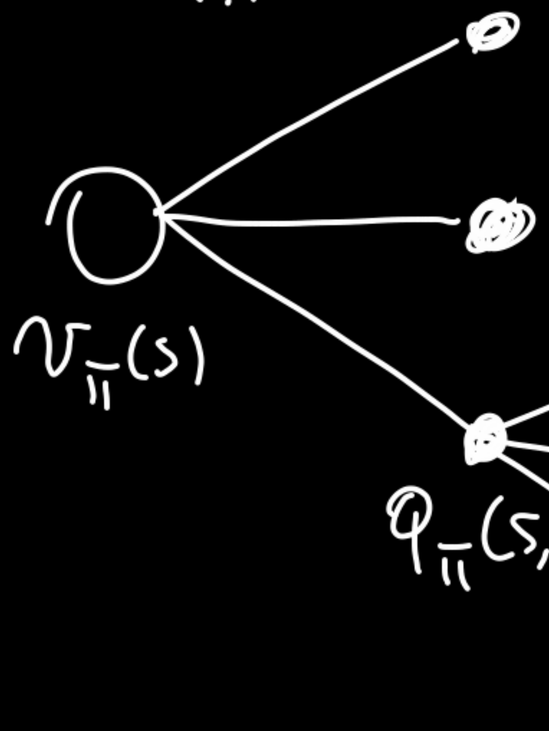
$$v_\pi(s) = \sum_a \underbrace{\pi(a|s)}_{\text{KNOWN}} \cdot \sum_{(s',r)} \underbrace{p(s',r|s,a)}_{\text{KNOWN}} \cdot \left[r + \gamma v_\pi(s') \right] \quad \forall s \ (n \text{ STATES})$$

RELATION BETWEEN VALUE OF A STATE AND VALUES OF SUCCESSOR STATES. MUST BE VALID FOR EVERY VALUE FUNCTION.

TRY TO FIND THE BELLMAN EQUATION FOR $q_\pi(s,a)$.

$$G_t = R_{t+1} + \underbrace{\gamma v_\pi(s')}_{r}$$

AGENT



$$v_\pi(s) = \sum_a \pi(a|s) \cdot q_\pi(s,a)$$

RELATION BETWEEN $v_\pi(s)$ AND $q_\pi(s,a)$

B.E.

BELLMAN EQUATIONS ARE m LINEAR EQUATIONS (m NUMBER OF STATES) IN m VARIABLES $\left(v_\pi(s_1), v_\pi(s_2), \dots, v_\pi(s_m)\right)$.

IN PRINCIPLE, GIVEN A POLICY $\pi(a|s)$, AND ASSUMING THE ENVIRONMENT IS KNOWN, WE CAN SOLVE B.E. AND FIND THE VALUE FUNCTION ASSOCIATED TO THE POLICY.

PROBLEM: IN MANY PRACTICAL PROBLEMS WE HAVE TOO MANY STATES FOR B.E. TO BE SOLVABLE IN PRACTICE.

↳ PRACTICAL SOLUTION.

OPTIMAL POLICIES

FOR FINITE MDP, WE HAVE A FINITE NUMBER OF DET. POLICIES.

IN GENERAL, THE SET OF POLICIES IS **PARTIALLY ORDERED**.

WITH VALUE FUNCTIONS, WE CAN JUDGE HOW GOOD POLICIES ARE.

π' IS BETTER THAN π ($\pi' \geq \pi$) IFF $v_{\pi'}(s) \geq v_{\pi}(s)$, $\forall s$.

(ON AVERAGE, FOLLOWING π' LEADS TO A HIGHER RETURN, $\forall s$)

OPTIMAL POLICY.

π^* IS OPTIMAL IFF $\quad v_{\pi^*}(s) \geq v_\pi(s), \; \forall s, \pi$

(OPTIMAL POLICY DOES BETTER - OR EQUALLY GOOD - THAN ANY OTHER POLICY).

- THERE IS AT LEAST ONE OPTIMAL POLICY,

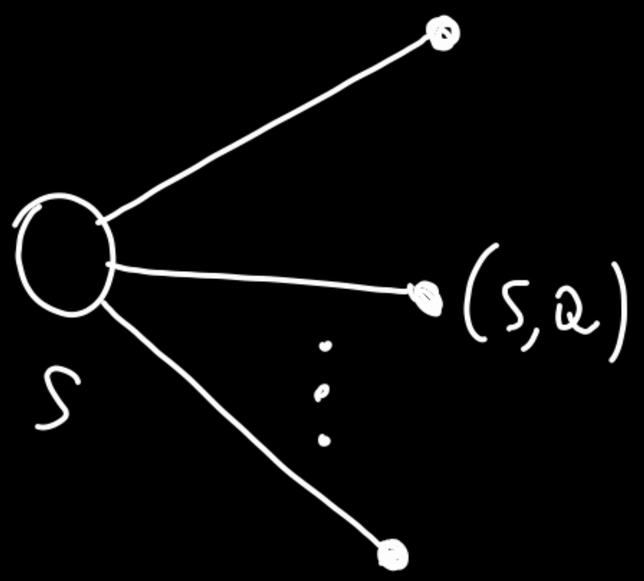
$v^*(s)$ AND $q^*(s,a)$ ARE THE (OPTIMAL) VALUE FUNCTIONS OF π^*.

$$\begin{cases} v^*(s) = \max_{\pi} \{ v_\pi(s) \}, \ \forall s \\ \\ q^*(s,a) = \max_{\pi} \{ q_\pi(s,a) \}, \ \forall s, a \end{cases}$$

$\pi(a|s)$

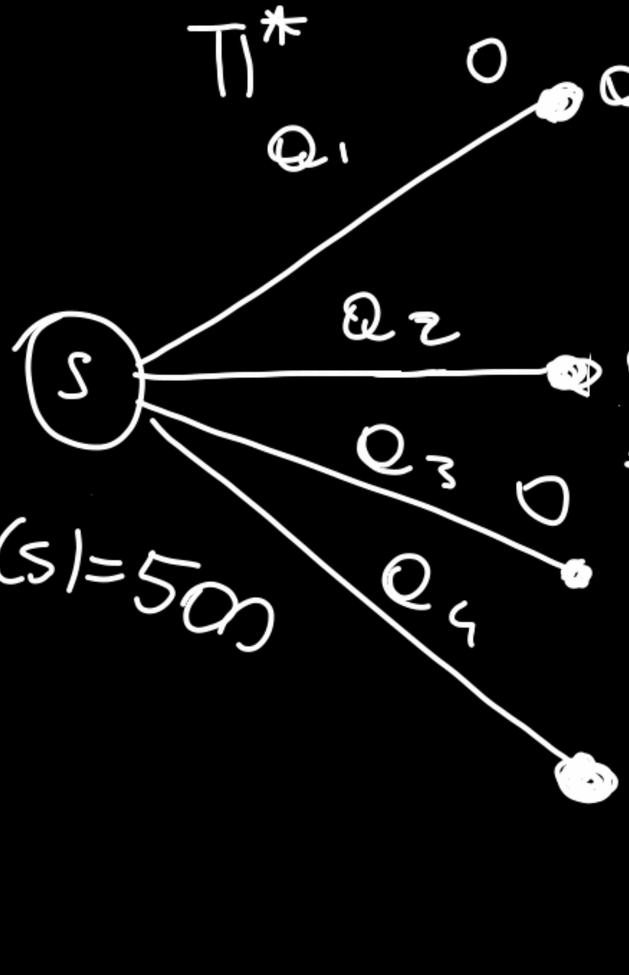
$$\boxed{v_\pi(s) = \sum_a \pi(a|s) \cdot q(s,a)}, \ \forall s.$$

OPTIMAL POLICY π^*



? NEXT SLIDE

OPTIMAL POLICY. RELATION BETWEEN $v^*(s)$, $q^*(s,a)$, π^*



$$\pi^* = \arg\max_a \{q^*(s,a)\}$$

$v^*(s)=500$

ACTION THAT MAXIMIZES EXPECTED FUTURE OPTIMAL RETURNS

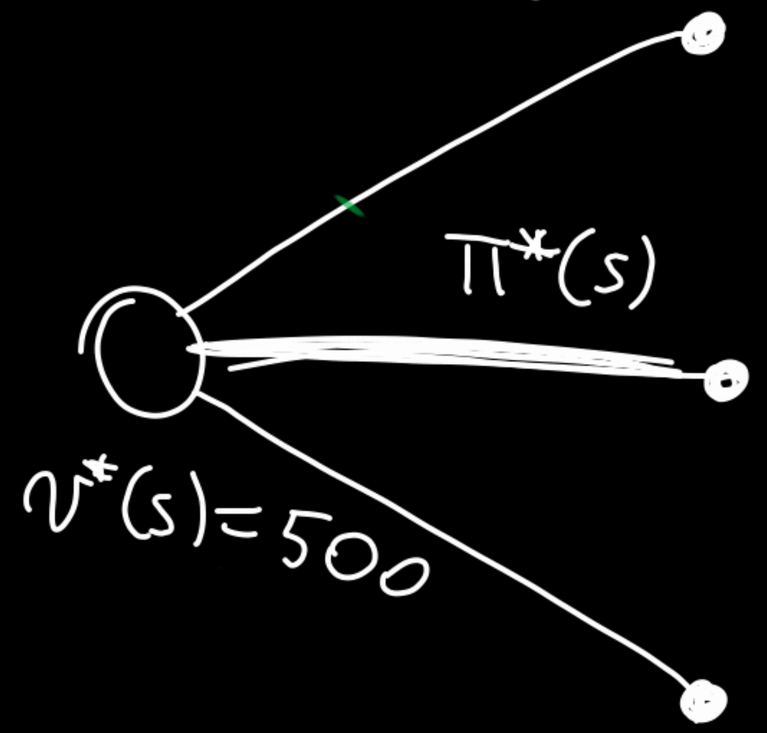
OPTIMAL POLICY IS GREEDY WRT q^*.

OPTIMAL VALUE FUNCTIONS

$$\pi^* = \arg\max_a \{ q^*(s,a) \}$$

GREEDY

$$v^*(s) = \max_a \{ q^*(s,a) \}$$



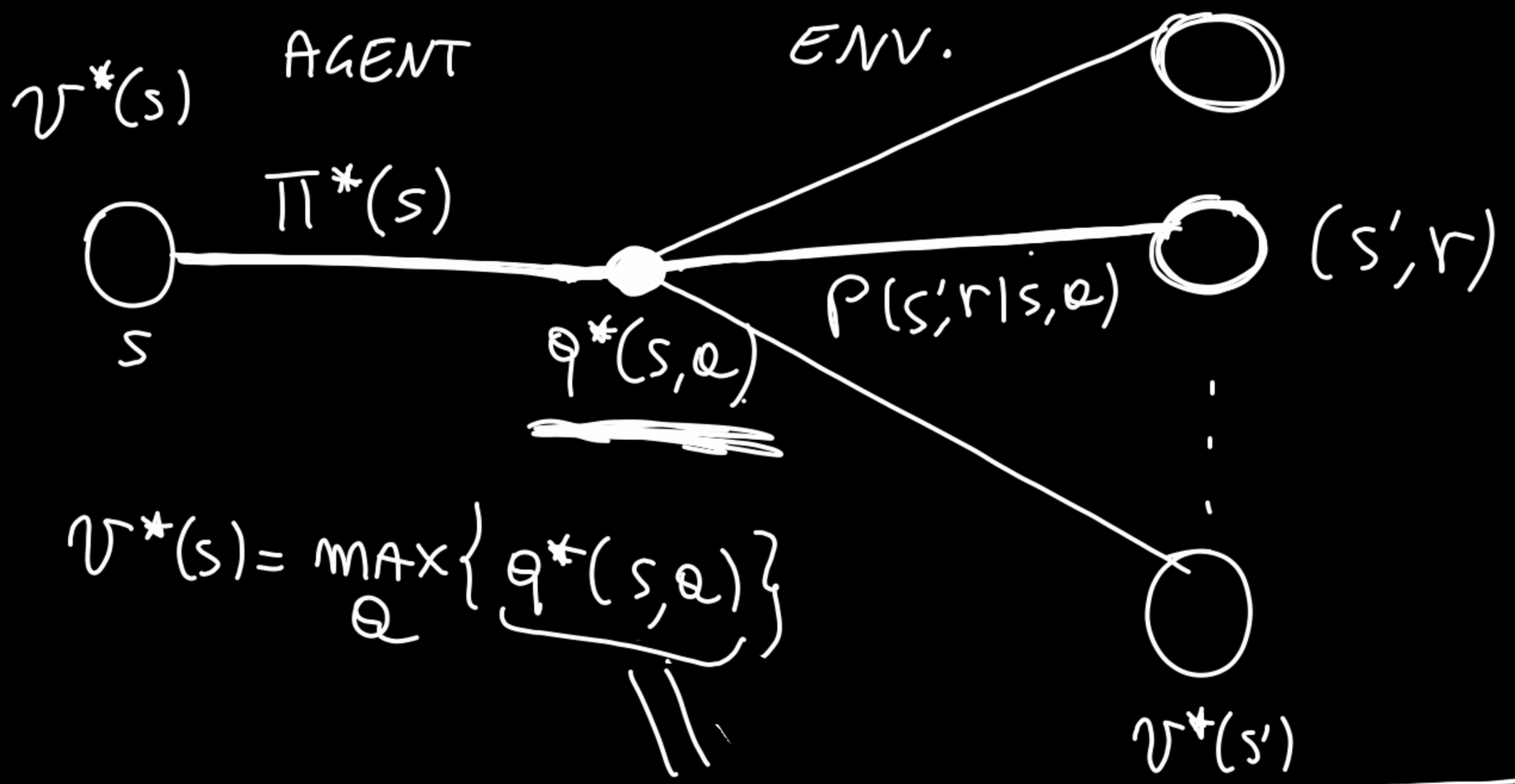
STATE NODE

STATE-ACTION NODE

FOR GENERAL POLICY

$$v_\pi(s) = \sum_a \pi(a|s) \cdot q_\pi(s,a)$$

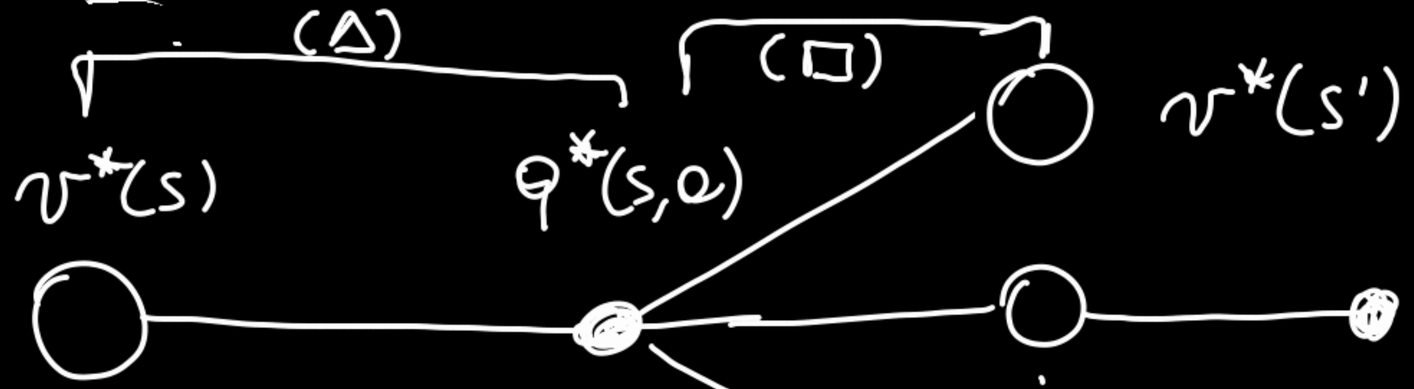
BELLMAN EQ. FOR v^* (OPTIMAL VALUE FUNCTION)



$$v^*(s) = \max_a \{ q^*(s,a) \}$$

$$v^*(s) = \max_a \left\{ \sum_{s',r} p(s',r|s,a) \left[r + \gamma v^*(s') \right] \right\}$$

RELATION BETWEEN $q^*(s,a)$ AND $v^*(s')$



(△)

$\pi^*(s) = \underset{a}{\operatorname{argmax}} \{ q^*(s,a) \}$, $\forall s$. ((OPTIMAL, GREEDY DETERM. POLICY)

$v^*(s) = \max_a \{ q^*(s,a) \}$

$v^*(s') = \max_{a'} \{ q^*(s',a') \}$

(□)

$q^*(s,a) = \sum_{s',r} p(s',r|s,a) \cdot [r + \gamma v^*(s')]$, $\forall (s,a)$ ← BELLMAN EQ. FOR q^*

$v^*(s) = \max_a \left\{ \sum_{s',r} p(s',r|s,a) [r + \gamma v^*(s')] \right\}$ ← BELLMAN EQ. FOR $v^*(s)$; $\forall s$

B.E. FOR v^* SET OF m (m STATES) NONLINEAR EQ. IN THE m VARIABLES $v^*(s_1), v^*(s_2), \ldots, v^*(s_m)$.

$$q^*(s,a) \longrightarrow \pi^* = \underset{a}{\operatorname{argmax}} \{ q^*(s,a) \}$$

$$\pi_0 \xrightarrow{\text{(EVALUATION)}} (v_{\pi_0}, q_{\pi_0}) \xrightarrow{\text{POLICY IMPR.}} \pi_1 \longrightarrow \cdots \pi^*.$$

(SOLVE B.E. / DIFFICULT)

$(\pi_1 \geq \pi_0)$